

Procured **AIR 1** in ACET; cleared CM1 (Maths), CS1 (Stats), CB1 (Finance), CB2 (Economics), CB3 (Management) from IAI
Appearing for the Chartered Financial Analyst (CFA) **Level 1** Exam in August 2025 for a rigorous foundation in finance

SCHOLASTIC ACHIEVEMENTS

- Awarded the prestigious Institute Academic Prize I in 2022–23 among **21** and 2023–24 among **20** from 1,400+
- Pursuing a **Minor** at the Centre for Machine Intelligence and Data Science; also achieved **10 SPI** in 8th semester
- Earned **Branch Change** to Industrial Engineering and Operations Research M.Tech. as one of the top **2** selected
- Scored **aaa.aa percentile** in JEE Mains and **aaa.aa percentile** in JEE Advanced among aaa million candidates
- Received **₹4 lakhs** INSPIRE scholarship for top **1%** in 12th and ranked **705** globally in IMO (114+ countries)
- Secured **99.93 percentile** in Mathematics in the Talent Search Test 2018 among more than 1 lakh candidates

PROFESSIONAL EXPERIENCE

Ernst and Young | *Actuarial Intern* (Jan '25-Mar '25)

- Implemented a robust **reserve** calculation strategy for Accidental Death Benefit, incorporating variable factors
- Improved efficiency **7.2×** and cut storage **96%** via Dask parallel processing and category compression in pandas
- Scaled processing from **1** million rows in Excel to over **20** million in Python for large-scale actuarial analysis

DT Portfolio Managers | *Financial Analyst* (May '25-June '25)

- Analyzed alkaline battery segment growth from **premiumization**, developed market trends and investor calls
- Boosted financial analysis **2.4×** and cut computation by **54%** by feeding zoomed images over PDFs to Perplexity
- Saved **5+** hours every month by automating portfolio screening using PyAutoGUI on the Bloomberg platform

FidelFolio Investment Advisory | *Quant Investment Analyst* (Feb '25-Apr '25)

- Revamped strategy analysis by back testing company fundamentals with **15+** advanced analytical visualizations
- Efficiently managed **missing** values for buy-hold and rebalance strategies, alongside sector distribution analysis
- Computed Internal Rate of Return with **PyXIRR**, using investor net worth & product returns to model cash flows

KEY PROJECTS

Scheduling Combat Logistics Force | *Indian Navy* (Dec '24-Present)

- Understood combat ship path needs and ideated algorithms for efficient tanker ship **scheduling** and **routing**
- Integrated **CBC** solver with a custom **heuristic**, addressing complex commodity constraints in naval operations
- Extended horizon to **27** days, enabling **₹400+** crore annual savings for Indian Navy through code optimization

Modelling Variance in Whole Life Insurance | *Markov Decision Processes* (Sep '24-Present)

Jay Vora | Full length paper under peer-review in Arab Journal of Mathematical Sciences (Emerald), Scopus/DOAJ-indexed

- Crafted a robust and systematic procedure to assess variance in whole life insurance policies under **constant** demand
- Modeled policies as Infinite Horizon Discrete-Time Discounted **Markov Chain** with exponential survival age
- Discovered that **demand** increases variance for policies with fewer initial policyholders compared to without it

Black - Scholes Model | *Course Project* | *Financial Econometrics* (Jan '25-Apr '25)

- Derived the Black-Scholes model for European options, demonstrating proficiency in financial derivatives valuation
- Tested and identified key limitations of the Black-Scholes model, including the **volatility smile** and **fat tails**
- Used Itô's lemma and geometric Brownian motion to build a riskless **hedge** portfolio, showing grasp of stochasticity

Monte Carlo Simulation M/M/2 queue | *Course Project* | *Modeling and Computation* (Aug '24-Sep '24)

- Developed MATLAB algorithms to efficiently simulate M/M/2 queue with **exponential** arrival and service times
- Gained expertise in functions like `exprnd()`, `poissrnd()`, `binornd()`, and `normrnd()` for sample generation
- Calculated average customer count, busy periods, and busy fraction for various **load factors** & initial condition

Team Shunya | *Junior Design Engineer* (Aug '22-May '23)

Earned 1st rank in 2 contests, 2nd in 2, 3rd in 3, and secured **2nd globally** in USSD called Olympics of Sustainable Architecture

- Persuaded **2** sponsors from **5** prospects and acted as a key contact for a **gold sponsor** during the project launch
- Optimized hot water and desiccant system with **cost-effective** pipe selection, sensor/valve setup and insulation
- Engineered a practical liquid desiccant setup for dehumidification and designed **2 HVAC** systems with AutoCAD

Reinforcement Learning (SoS) | *Mathematics & Physics Club*

(May '22-July '22)

- Designed code for the Cart Pole problem to run it for **500** frames using **os**, **gym** and **stable_baselines3** libraries
- Learned **Finite-State Markov Chains**, Markov Decision Processes, Upper Confidence Bound Action Selection, Action-Value Methods, ϵ -greedy method, Optimistic Initial Values and efficiently handling non-stationary problems

Multilayer Perceptron Implementation | *Course Project | Machine Learning*

(Mar '23-Apr '23)

- Pioneered and collaborated in implementing an advanced **MLP model** with a potent sigmoid activation function
- Harnessed the **MNIST** dataset and **TensorFlow**'s prowess to transform images into robust numerical arrays
- Achieved a stellar accuracy of **100%** and **99.92%** for numerical data and image classification, respectively

Dynamic Vehicle Routing Problem | *Course Project | Optimization Methods in Engineering*

- Built a MIP model in Python for **dynamic** vehicle routing, minimizing delivery costs via optimal locations and times
- Developed a model accounting for delivery, handling, **holding costs** with capacity and parcel weight constraints
- Implemented with CBC solver and validated on both small and large cases for logical, **efficient** delivery solutions

Social Network Influence | *Course Project | Applied Integer Programming*

(Jan '25-Apr '25)

- Designed a **bi-objective** model minimizing seed nodes while maximizing expected influenced nodes across scenarios
- Applied **Pareto Frontier** for optimal solutions, balancing seed costs and variance as a penalty in objective value

Last-Mile Delivery Simulation | *Course Project | Simulation Modeling*

(Jan '25-Apr '25)

- Modeled discrete-event simulation of last-mile delivery system in AnyLogic for **44** days, handling **994** orders daily
- Analyzed KPIs, observing delivery agent utilization of **99.9%** and a **138.2**-minute average delivery process time

Strategizing Location of Wireless Network | *Course Project | Operations Modeling*

(Jan '24-May '24)

- Factored wall impact on signal strength to determine optimal router locations in complex layouts, using python
- Utilized **Voronoi** diagrams, Welzl's algorithm, and modified Univariate search while considering corner cases

Inverse Projection 2D to 3D | *Course Project | Computer Graphics and Product Modeling*

(Aug '24-Nov '24)

- Created an algorithm using positive/negative extrusions to rebuild partially convex **3D** models from **2D** views
- Developed a method to tackle complex cases by detecting **cutouts** and deriving dimensions from projections

Sustainable Solar Tracking Mechanism | *Course Project | Machine Design*

(Aug '24-Nov '24)

- Designed a solar tracker with feed-forward control, optimizing energy capture without energy-intensive sensors
- Modeled key design parameters, including a 1 kg actuator and 300 N/m spring, achieving **-20°** to **20°** rotation
- Validated prototype performance, showing over **20%** increase with 8505 Wh/m² vs 8083 Wh/m² for fixed panel

Fabricating Compressor | *Course Project | Fluid Mechanics*

(Aug '22-Nov '22)

- Led a dedicated team of **3** to build a compressor with maximum pressure **402 Pa** and an outlet velocity of **15 cm/s**
- Designed the piston using an **in-line slider-crank** mechanism having 0 eccentricity to ensure smooth motion
- Optimized the length of the rod to **15 cm** by carefully accounting for the **buckling** effect and **force** maximization

Solid vs Hollow Cylinders: Differences in Properties | *Course Project | Solid Mechanics*

(Mar '23-Apr '23)

- Innovated approach to measure torsional properties of **diverse** aluminum tubes with systematic evaluation
- Validated moment of area's role on limiting torque with **6.86%** error, giving insights on asymmetric torque
- Concluded larger-diameter hollow tubes deliver superior torque, with implications for high-load applications

LEADERSHIP EXPERIENCE

Stock Prediction using Deep Reinforcement Learning (WiDS) | *Analytics Club*

(Dec '23-Jan '24)

- Spearheaded the mentorship of **9** students, cultivating their proficiency in finance and reinforcement learning
- Built RL models with gym-anytrading and crafted support resistance, MACD and ADX indicators collaboratively

Momentum Based Portfolio Strategy (FinSearch) | *Finance Club*

(June '23-Sep '23)

- Mentored a cohort of aspiring students, facilitating their journey through expert guidance and curated resources
- Proactively oversaw project progress, addressed inquiries, reviewed reports, and polished final presentations

Interview Coordinator | *Placement Cell*

(Oct '23-Dec '23)

- Orchestrated the effective administration of placement tests for **20+** companies while handling student queries
- Coordinated closely with a dynamic team of **250+** members to successfully manage interviews for **2300+** students

Natural Language Processing (SoS) | *Mathematics & Physics Club*

(May '23-Aug '23)

- Mentored **4** students, introducing tokenization, stemming, and word embeddings providing project guidance
- Tracked progress by resolving technical queries and reviewing reports on language models and semantic analysis

EXTRACURRICULAR

- Represented Gujarat state **3 times** as **captain** and proudly garnered **1 silver** medal at the national level in Yoga
- Gained **2 gold**, **7 silver** and **4 bronze** medals in state; **4 gold**, **3 silver** and **1 bronze** medals in district in Yoga
- Assisted **120+** invited guests at Pramukh Swami Maharaj Shatabdi Mahotsav having a daily footfall of **3.9** lakh
- Clinched **silver** medal in chess and another **silver** medal in drawing at the district level, showcasing versatility
- Teaching Assistant MDP ['25] | Stack Exchange ['21, 75 badges] | Wikipedia contributor ['21] | NSS SSD ['21-22]
- Technical Skills:** Python, SQL, R, Spark, MATLAB, Power BI, Jira, VBA, Bloomberg, C++, MS Excel, MathJax, Canva